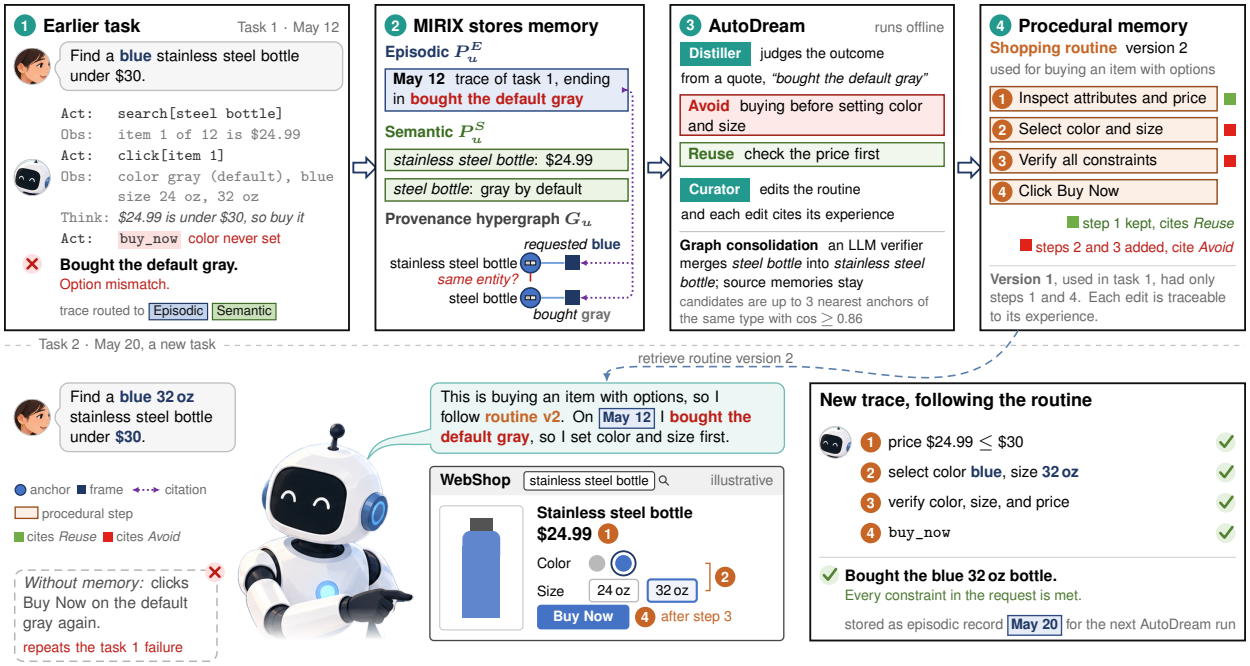
1 Earlier task
Task 1 · May 12
Find a blue stainless steel bottle under $30.
Act: search[steel bottle]
Obs: item 1 of 12 is $24.99
Act: click[item 1]
Obs: color gray (default), blue size 24 oz, 32 oz
Think: $24.99 is under $30, so buy it
Act: buy_now color never set
Bought the default gray.
Option mismatch.
trace routed to Episodic Semantic
2 MIRIX stores memory
Episodic P_u^E
May 12 trace of task 1, ending in bought the default gray
Semantic P_u^S
stainless steel bottle: $24.99
steel bottle: gray by default
Provenance hypergraph G_u
requested blue
stainless steel bottle
same entity?
steel bottle
bought gray
3 AutoDream
runs offline
Distiller judges the outcome
from a quote, "bought the default gray"
Avoid buying before setting color and size
Reuse check the price first
Curator edits the routine
and each edit cites its experience
Graph consolidation an LLM verifier merges steel bottle into stainless steel bottle; source memories stay
candidates are up to 3 nearest anchors of the same type with cos ≥ 0.86
4 Procedural memory
Shopping routine version 2
used for buying an item with options
1 Inspect attributes and price
2 Select color and size
3 Verify all constraints
4 Click Buy Now
step 1 kept, cites Reuse
steps 2 and 3 added, cite Avoid
Version 1, used in task 1, had only steps 1 and 4. Each edit is traceable to its experience.
Task 2 · May 20, a new task
retrieve routine version 2
Find a blue 32 oz stainless steel bottle under $30.
This is buying an item with options, so I follow routine v2. On May 12 I bought the default gray, so I set color and size first.
anchor frame citation
procedural step
cites Reuse cites Avoid
Without memory: clicks Buy Now on the default gray again.
repeats the task 1 failure
WebShop
stainless steel bottle
illustrative
Stainless steel bottle
$24.99 1
Color
Size
24 oz
32 oz
2
Buy Now
4 after step 3
New trace, following the routine
1 price $24.99 ≤ $30
2 select color blue, size 32 oz
3 verify color, size, and price
4 buy_now
Bought the blue 32 oz bottle.
Every constraint in the request is met.
stored as episodic record May 20 for the next AutoDream run